# agriBORA Commodity Price Prediction Challenge

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**Competition:** agriBORA Commodity Price Prediction Challenge (Zindi)

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## 1. Executive Summary

The objective of this project was to develop a machine learning model capable of forecasting the wholesale prices of **Dry White Maize** in five specific Kenyan counties (Kiambu, Kirinyaga, Mombasa, Nairobi, and Uasin-Gishu) for Week 52 of 2025 and Week 1 of 2026. The solution employs a Gradient Boosting Regressor (CatBoost) that integrates historical price data with external market indicators to produce high-accuracy time-series forecasts.

## 2. Methodology and Data Preparation

### 2.1. Data Integration

To ensure the model captured the most recent price volatility, multiple datasets were merged:

- **Primary Data**: Historical maize prices from agriBORA_maize_prices.csv.
- **Supplementary Data**: Recent price updates for weeks 46 to 51 (agriBORA_maize_prices_weeks_46_to_51.csv) were appended to the training set.
- **External Indicators**: Market data from KAMIS (kamis_maize_prices.csv) was utilized to extract average wholesale, retail, and supply volume trends per county.

### 2.2. Feature Engineering

Transforming raw data into meaningful features was critical for model performance. The following features were engineered:

- **Log Transformation**: The target variable (WholeSale) was converted to log_price to stabilize variance and normalize the distribution.
- **Lag Features**: Lags of 1, 2, 3, and 4 weeks were created to capture short-term historical dependencies.
- **Rolling Statistics**: 3-week and 5-week moving averages were calculated to smooth out noise and identify local trends.
- **Trend Indicator**: A custom feature was derived by calculating the difference between the 5-week rolling average and the previous week's price.
- **Seasonality**: Month and WeekofYear were extracted to account for seasonal agricultural cycles.

# 3. Modeling and Training

## 3.1. Algorithm Selection

The **CatBoostRegressor** was selected due to its superior handling of categorical variables (County IDs) and its robustness against overfitting in relatively small datasets.

## 3.2. Hyperparameters

The model was trained with the following configurations to achieve the best RMSE (Root Mean Squared Error):

- **Iterations**: 1200
- **Learning Rate**: 0.05
- **Depth**: 8
- **Loss Function**: RMSE
- **Random Seed**: 42 (for reproducibility)

# 4. Inference Strategy

For the final submission, the model identified the **Anchor Week** (the most recent week where data was available for all target counties). The model then predicted the prices for the target

periods by utilizing the state of features from that anchor week. The final predictions were converted back from the log-scale using the exponential function (np.exp).

# 5. How to Reproduce Results

### System Requirements

- Python 3.x
- Required Libraries: catboost, pandas, numpy, scikit-learn

### Execution Steps

1. **Data Placement**: Ensure all source CSV files (agriBORA and KAMIS datasets) are placed in the directory defined by the DATA_DIR variable in the code.
2. **Run Pipeline**: Execute the cells in the provided notebook (FINAL_SUBMISSION_WEEK52_WEEK1.ipynb) in sequential order.
3. **Output**: The script will automatically generate a file named FINAL_SUBMISSION_WEEK52_WEEK1.csv containing the predicted values for each County and Week combination.

# 6. Conclusion

This methodology effectively combines historical price lags with regional market trends. By leveraging CatBoost's powerful gradient boosting architecture, the solution provides a stable and reliable forecast for maize prices, meeting the requirements of the agriBORA Commodity Price Prediction Challenge.